06. Model Selection

Understand the two clustering algorithms and the role of PCA and K selection.

Project: SmartCart Customer Segmentation

Source checked: Sai-Likhith-2005/Customer-Segmentation | smartcart.ipynb | smartcart_customers.csv

K-Means

K-Means partitions observations into K groups by minimising within-cluster sum of squared Euclidean distances (WCSS): sum over clusters and members of the squared distance from each point to its cluster centroid. It is efficient and produces centroids, but favours roughly spherical clusters, is sensitive to scaling and outliers, and requires K in advance.

Agglomerative hierarchical clustering

Agglomerative clustering begins with each observation as its own cluster and repeatedly merges the closest groups. The notebook uses Ward linkage, which merges clusters to minimise the increase in within-cluster variance. It produces a hierarchy, but fitting all records can be more computationally expensive and standard scikit-learn AgglomerativeClustering does not provide a native predict method for new customers.

PCA

PCA is dimensionality reduction, not a clustering algorithm. The notebook uses three principal components as the space in which it runs K-Means and Agglomerative clustering. The first three components explain 44.95% of standardized feature variance.

Choosing K

- Elbow method: inspect WCSS as K increases and look for diminishing returns. The notebook uses KneeLocator and reports K=4.
- Silhouette: compares a point's average distance to its own cluster with its average distance to the nearest other cluster. Values range from -1 to 1; higher is generally better.
- No single metric determines the commercially correct K. Compare interpretability, stability, segment size and actionability as well.

What the repository actually does

Stage	Implementation
K exploration	K-Means for K=1..10; random_state=42; stores inertia
Elbow result	KneeLocator reports optimal_k = 4
Silhouette	K-Means for K=2..10; plots scores; numeric scores not printed
Final K-Means	KMeans(n_clusters=4, random_state=42)
Final labels used for profile	AgglomerativeClustering(n_clusters=4, linkage='ward')
PCA input	3-component PCA representation

Why two algorithms appear

The notebook uses K-Means to explore K and generate a visualisation, then uses AgglomerativeClustering to create the final labels for the cluster-characterisation table. These are separate partitions; do not describe the final customer labels as K-Means labels. The final profile labels are from Agglomerative clustering.

Alternatives

- K-Medoids: more robust representatives than means, if available.
- Gaussian Mixture Models: soft membership probabilities and elliptical clusters.

- DBSCAN/HDBSCAN: density-based clusters and noise points; may struggle with varying density and high-dimensional scaled data.
- Business-rule segmentation: useful as a transparent baseline, even if not a purely unsupervised method.